MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER Discipline Specific Elective (DSE) - MG4DSEGEO202 - GROUNDWATER HYDROLOGY

ANSWER SCHEME

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

(Remember(K), Understand(U), Apply(A), Analyse (An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation (Ap))

Part A: Short Answer (60 words)

Answer any 5. Each carries 2 marks.

1. Define vertical distribution of groundwater. [K] / [CO1]

- Division of subsurface water into two main zones: the **Zone of Aeration** (Vadose zone) and the **Zone of Saturation** (Phreatic zone). [1 Mark]
- Separated by the **Water Table**; describes how water occupies voids from the surface down to the impermeable bed. [1 Mark]

2. State the role of GIS in remote sensing. [K] / [CO2]

- GIS acts as a tool for **storing and analysing** spatial data acquired via remote sensing. [1 Mark]
- Allows for **overlay analysis** (e.g., combining geology and land use maps) to identify potential groundwater zones. [1 Mark]

3. List two chemical parameters for drinking water. [K] / [CO3]

- Any chemical parameters such as - **pH value** (identifies acidity/alkalinity). [1 Mark]
- **Total Dissolved Solids (TDS)** or **Fluoride/Chloride concentration**. [1 Mark]

4. Classify groundwater contaminants. [U] / [CO4]

- Definition of groundwater contaminants. [1 Mark]
- **Natural/Geogenic:** (e.g., Arsenic, Fluoride leaching from rocks). [0.5 Mark]
- **Anthropogenic:** (e.g., Nitrates from fertilizers, industrial effluents, or pathogens from sewage). [0.5 Mark]

5. State two limitations of artificial recharge. [K] / [CO4]

- **Clogging of aquifers (siltation):** Silt or microbial growth can reduce permeability at the injection site. [1 Mark]
- High initial investment for infrastructure and maintenance. [1 Mark]

6. List the main components of the hydrologic cycle. [K] / [CO1]

- Evaporation/Evapotranspiration and Precipitation. [1 Mark]
- Infiltration, Runoff, and Groundwater flow. [1 Mark]

Part B: Short Notes (250 words)

Answer any 6. Each carries 8 marks.

7. Describe SP method in the context of surface geophysical method. [U] / [CO2]

- **Principle:** Measuring natural potential differences (Self-Potential) in the earth without an artificial current. [2 Marks]
- **Mechanism:** Potentials arise from electrochemical activity or "streaming potential" (water flowing through porous media). [3 Marks]
- **Application:** Useful for locating ore bodies, detecting leaks in dams, or mapping shallow groundwater flow. [3 Marks]

8. Discuss physical, chemical, and bacteriological standards (WHO & BIS). [U] / [CO3]

- **Physical:** Colour, Odor, turbidity, and taste standards (e.g., Turbidity < 5 NTU). [2 Marks]
- **Chemical:** Limits for pH (6.5-8.5), Hardness, Iron, and Nitrates. Comparison of BIS (IS 10500) vs. WHO guidelines. [3 Marks]
- **Bacteriological:** Requirement for zero *E. coli* or coliform counts in 100ml samples for safe potable water. [3 Marks]

9. Groundwater overexploitation, hydraulic gradient, and saline water intrusion. [U] / [CO4]

- **Mechanism:** Over-pumping lowers the water table, creating a "cone of depression." [2 Marks]
- **Hydraulic Gradient:** Reversal of the gradient causes seawater to flow inland toward the well. [2 Marks]
- **Saline Intrusion:** Ghyben-Herzberg relation (freshwater-saltwater interface). [2 Marks]
- **Prevention:** Controlled pumping, artificial recharge, and creating hydraulic barriers. [2 Marks]

10. Variations in groundwater quality across Indian provinces. [U] / [CO5]

- **Arsenic:** Dominant in West Bengal and Ganga-Brahmaputra plains. [2 Marks]
- **Fluoride:** High in arid/semi-arid regions like Rajasthan, Andhra Pradesh, and parts of Kerala (Palakkad). [2 Marks]
- **Salinity:** Coastal regions (Gujarat/Tamil Nadu) and inland salinity in Punjab/Haryana due to irrigation. [2 Marks]
- **Iron:** Common in North-Eastern states and parts of Kerala/Odisha. [2 Marks]

11. Illustrate how secondary porosity affects water occurrence in hard rocks. [U] / [CO1]

- **Definition:** Porosity created after rock formation via fracturing, jointing, or weathering. [2 Marks]
- **Crystalline Rocks:** Igneous/Metamorphic rocks have zero primary porosity; water is stored only in fractures. [3 Marks]
- **Occurrence:** Highly localized; yields depend on fracture connectivity and the thickness of the weathered "saprolite" zone. [3 Marks]

12. Briefly describe various types of sub-surface geophysical methods. [U] / [CO2]

- **Borehole Logging:** Electrical Resistivity logs (Normal/Lateral) to identify aquifers. [3 Marks]
- **Gamma Logging:** Identifying clay layers vs. sand. [2 Marks]
- **Sonic/Neutron logging:** Measuring porosity and density within the drilled hole. [3 Marks]

13. Explain how high SAR values cause soil dispersion and reduced permeability. [U] / [CO3]

- **Definition:** Sodium Adsorption Ratio (SAR) measures the ratio of Na^+ to Ca^{2+} and Mg^{2+}. [2 Marks]
- **Process:** High Sodium replaces Calcium in clay, causing clay particles to swell and disperse. [3 Marks]
- **Impact:** Dispersed particles block soil pores, destroying soil structure and drastically reducing permeability/infiltration. [3 Marks]

14. Radioactive contamination in groundwater. [U] / [CO4]

- **Sources:** Natural (Radon/Uranium leaching from granitic rocks) and Anthropogenic (Mining, nuclear waste). [3 Marks]
- **Factors:** Geological parent material, pH (solubility), and redox conditions of the aquifer. [3 Marks]
- **Distribution:** High concentrations often found in deep fractured aquifers or near mineralized zones. [2 Marks]

Part C: Essays (400 words)

Answer any 1. Carries 12 marks.

15. Categorize groundwater basins and explain hydrological boundaries. [U] / [CO1]

- **Introduction:** Definition of a groundwater basin as a hydrogeologic unit containing one or more aquifers. [2 Marks]
- **Categories:** Open vs. Closed basins; Rock-bounded vs. Alluvial basins. [4 Marks]
- **Hydrological Boundaries:**
    - **Physical:** Geological contacts, faults, or impermeable strata. [2 Marks]

    - **Hydraulic:** Groundwater divides (crests in water table), streamline boundaries, and constant head boundaries (large rivers/oceans). [4 Marks]

16. Process of liquid waste disposal and groundwater degradation. [U] / [CO4]

- **Disposal Methods:** Septic tanks, rapid infiltration pits, deep well injection, and lagooning. [4 Marks]
- **Degradation Process:** Leaching of pollutants through the vadose zone; formation of a "Contaminant Plume." [4 Marks]
- **Impacts:** Nutrient enrichment (Nitrates), Pathogen introduction, Heavy metal accumulation, and change in pH/Redox chemistry which can trigger further mineral leaching. [4 Marks]